

The Label Provenance Problem in DFU Severity Grading: Validating Tissue-Derived Labels, Leakage-Free Evaluation, and Tissue-Mapped Explainability

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Abstract—No public diabetic foot ulcer dataset carries severity annotation, so recent work derives severity labels from image content. This paper implements one such derivation, clustering wound pixels by colour and thresholding the necrotic fraction, and then subjects the resulting labels to four independent validation checks rather than assuming they are correct. All four fail. Agreement with two clinicians is indistinguishable from chance ($\kappa \approx 0$), against an inter-rater agreement of $\kappa = 0.25$ that bounds what any automated method could achieve. A negative control returns a severe grade for 60.2% of lesion-free skin images. We identify the mechanism: k -means with $k = 3$ cannot express the absence of a tissue type, so the darkest region of any image is labelled necrotic by construction, including shadow, wound depth and darker skin. To separate label quality from pipeline quality, the identical architecture and protocol are applied to an expert-annotated infection task, used strictly as a control and not as a substitute for severity, where the same code reaches AUROC 0.81. Severity grading from colour wound images is therefore limited by the absence of clinical annotation rather than by model capacity, and we release the validation protocol that establishes this.

Index Terms—Diabetic foot ulcer, label validation, data leakage, negative control, wound image analysis, ordinal regression, explainable AI.

I. INTRODUCTION

DIABETES mellitus affects a large and growing share of the world population, and the International Diabetes Federation reported 537 million adults living with the disease in 2021, with projections reaching 783 million by 2045 [1]. Among its complications, diabetic foot ulcers (DFU) carry the heaviest clinical and economic burden. Between 25% and 34% of diabetic patients develop an ulcer during their lifetime [2], and more than 85% of diabetes-related lower-limb amputations are preceded by a foot ulcer [3]. In lower-middle-income countries the cost of treatment can reach several times a patient’s annual income [4].

Clinicians grade DFU severity using systems such as the Wagner scale [5] or the University of Texas classification [6], both of which map the wound to an ordinal category that determines the treatment pathway. Assessment is performed by visual inspection supported by physical probing, which introduces inter-clinician variability and is unavailable in settings without wound care specialists [7]. Automating this

assessment from a photograph would extend triage capacity to primary care and remote settings.

Deep learning has been applied to DFU image analysis since 2018 [8], and progress accelerated after the Diabetic Foot Ulcer Challenge series introduced standardised benchmarks [9]. A 2025 systematic review screening 4,653 articles found that machine learning models achieve accuracies between 0.88 and 0.97 on binary detection and on infection or ischaemia classification [10]. Severity grading, however, remains unsolved on public data. Studies that attempt Wagner grading rely on private hospital datasets that cannot be reproduced [11], [12], or collapse adjacent grades because labelled data is insufficient [11].

The obstacle is not model capacity but annotation. No publicly downloadable DFU dataset carries severity labels. A natural response, and the one this work initially pursued, is to derive severity from image content: segment the wound into tissue types by colour, compute the proportion of each, and apply threshold rules anchored to published grading criteria. The approach is attractive because it requires no clinician time and scales to any unlabelled corpus.

This paper reports what happens when that derivation is validated rather than assumed correct. It does not recover severity, and we document both the evidence and the mechanism. To establish that the failure lies in the labels and not in the surrounding pipeline, the same architectures and evaluation protocol are applied to a wound classification task that does carry expert annotation, where they perform as expected. The conclusion is therefore specific: severity grading from colour wound images is not blocked by model capacity or image resolution, but by the absence of clinically annotated ground truth.

A second finding emerged from the same experiments. During the study we designed TGO-Net, a tissue-guided architecture that injects the derived tissue proportions into a frozen convolutional representation through a learned multiplicative gate whose scalar is initialised at zero, so the network reports for itself how much of that signal it uses. On the derived labels the gated model appears to perform extremely well, but we show this is circular: the tissue proportions are the label-generating variable. On expert labels, where no such circularity exists, the gate never stabilises and the gated model performs slightly worse than a plain CNN. The architecture is retained in this paper as a diagnostic instrument rather than a performance

contribution, because the behaviour of its gate is itself evidence about the labels.

The contributions of this work are:

- A **colour-based tissue proportion labelling method** that derives mild, moderate and severe labels from wound images without ground-truth annotations, specified in full so that it can be reproduced and independently assessed.
- A **four-part validation protocol** for derived image labels, combining expert agreement, a negative control on lesion-free images, a stability test across augmented copies of the same source photograph, and a corpus plausibility check. Each check fails independently, and each is inexpensive relative to the cost of training on invalid labels.
- **Identification of the mechanism** behind the failure: k -means with $k = 3$ partitions every image into three non-empty clusters and therefore cannot express the absence of necrosis; the assumed luminance ordering of tissue types is inverted; and all dark image regions, including darker skin, are absorbed into the cluster labelled necrotic.
- A **leakage-free evaluation protocol** for pre-augmented corpora, using dihedral-invariant content hashing, photograph-level identity recovery from the union of filename and pixel evidence, and patient-grouped folds verified on the image table across five independent keys.
- **TGO-Net**, comprising a Tissue Proportion Encoder, a Cross-Modal Gate with a zero-initialised scalar, and a rank-consistent CORAL head, presented with its parameter budget and with the gate scalar used as an interpretable measure of modality reliance.
- A **control experiment** applying the identical architectures, preprocessing and evaluation protocol to an expert-annotated infection task, isolating label provenance as the cause of failure rather than model capacity, input resolution or training configuration.

The remainder of this paper is organised as follows. Section II reviews related work. Section III states the critical gaps in prior methods. Section IV lists the core contributions. Section V describes the methodology. Section VI details the datasets and experimental setup. Section VII reports results. Section VIII discusses them. Sections IX through XI state limitations, future work and conclusions.

II. RELATED WORK

A. Severity Grading from Wound Images

The most direct prior work on Wagner grading from images is Zhou et al. [11], who applied Mask2Former to segment seven tissue types and assign grades on 671 annotated images from a Chinese hospital, reaching AUC 0.94 on tissue segmentation. Grades 1 and 2 were collapsed and grades 0 and 5 excluded because of insufficient samples, and the dataset is not public. Abebe et al. [12] applied MobileNetV2 to full six-class Wagner grading on 3,700 images from three Ethiopian hospitals, reporting 82% validation accuracy, again on private data and without cross-dataset validation. Anilkumar et al. [13] compared six architectures on a 407-image South Indian

clinical dataset labelled by IWGDF criteria, with MobileNetV2 reaching 82% accuracy and adjacent-grade confusion identified as the dominant failure mode. Wang et al. [23] proposed ScoreDFUNet, classifying images into ulcer, infection, normal and gangrene zones on 1,426 images with 95.34% accuracy, without addressing the full severity scale.

The pattern across these studies is consistent: severity grading works when clinicians have graded the images, and every such dataset is private. None reports a protocol for deriving severity labels from public unlabelled data, and none validates such labels.

B. Architectures for DFU Classification

The standard architecture shifted from custom convolutional networks to pretrained backbones after the challenge series demonstrated that fine-tuned EfficientNet models outperform task-specific designs on infection and ischaemia classification [16]. Subsequent work has largely pursued attention mechanisms and feature fusion on the same benchmarks: transformer-attention hybrids [14], attention-driven dense networks, reconstruction residual networks with spatial-channel attention [17], handcrafted texture descriptors combined with learned embeddings [18], and patch-based networks with context-preserving attention [19]. Reported macro F1 on the DFUC-2021 infection and ischaemia task clusters around 80%.

These studies share a limitation relevant here: all evaluate on labels supplied with the dataset, and none reports whether augmented copies of the same photograph, or multiple wound sites from the same patient, were separated across training and test splits.

C. Explainability in DFU Models

Explainability work in this domain applies SHAP, LIME and Grad-CAM to binary detection, reporting high accuracies and heatmaps that localise the ulcer region [15], [20]. A recent review found that only 23% of DFU studies incorporate explainability methods at all, and that none maps activation outputs to clinical wound tissue features [21].

Across this literature, explainability outputs are generated at the detection level and are not linked to validated tissue references. Our analysis in Section VII-E addresses why doing so is harder than it appears when the tissue reference is itself derived by clustering.

D. Label Quality and Data Leakage in Medical Imaging

Concern about label quality and evaluation integrity is not specific to DFU. Reviews of machine learning in medical imaging repeatedly identify optimistic evaluation arising from improper data partitioning, in particular when multiple images originate from the same patient or the same acquisition [36], [22]. Gomes et al. [10] note that only 67.3% of DFU classification datasets are publicly accessible, which limits independent verification of both labels and splits. Dhar et al. [37] report a pilot study on wound tissue segmentation and observe that inter-annotator variation is substantial even among trained annotators, which is consistent with the inter-rater agreement we measure in Section VII-A.

To our knowledge no prior DFU study reports a negative control on lesion-free images, or a stability analysis across augmented copies of the same photograph, as a precondition for trusting derived labels.

III. CRITICAL GAPS AND LIMITATIONS OF PRIOR METHODS

Four gaps follow from the literature above, and the contributions in Section IV map onto them one to one.

Gap 1: derived labels are used without validation. Where public data lacks severity annotation, labels are derived from image content and then used directly for training. No prior study reports whether the derived labels agree with clinicians, whether they behave correctly on images containing no lesion, or whether they are stable under transformations that do not change the clinical picture. Contribution 2 supplies a validation protocol; contribution 3 reports the mechanism when it fails.

Gap 2: evaluation integrity on pre-augmented corpora is not reported. Public wound corpora are frequently distributed after augmentation, so a split over image files does not separate photographs. None of the reviewed studies states how augmented copies or per-patient multiplicity were handled. Contribution 4 supplies a protocol that addresses this and verifies it on the image table rather than on an aggregated table.

Gap 3: architectural additions are not tested against a label-quality confound. Where auxiliary features derived from the same images are added to a network, an improvement may reflect the auxiliary feature carrying label information rather than clinical signal. Contribution 5 introduces an architecture whose gate scalar makes modality reliance directly readable, and contribution 6 supplies the control condition needed to interpret it.

Gap 4: no control separates label quality from pipeline quality. When a method underperforms, the literature does not distinguish between inadequate labels and an inadequate model. Contribution 6 applies the identical pipeline to an expert-annotated task on comparable images, isolating the variable.

IV. CORE CONTRIBUTIONS

The contributions listed in Section I are developed as follows: the labelling method in Section V-B, the validation protocol in Section V-C with results in Section VII-A, the mechanism in Section VII-B, the evaluation protocol in Section V-D, TGO-Net in Sections V-E and V-F, and the control experiment in Section VII-C.

V. PROPOSED METHODOLOGY

The pipeline comprises seven stages: identity resolution, label derivation, label validation, split construction, feature extraction, model training with ablation, and statistical analysis. Each stage writes an artefact consumed by the next, and each asserts its preconditions before running, so that a stage cannot silently operate on incomplete input.

A. Pipeline Overview

Fig. 1 shows the end-to-end dataflow. Two corpora enter stage A, where every image receives a photograph identifier and a patient identifier. Stage B derives severity labels on the first corpus. Stage C validates those labels and consolidates them to one label per photograph. Stage D constructs patient-grouped folds and verifies that no identity key spans a fold boundary. Stages E and F train models on the derived labels and on expert labels respectively, using the same folds and the same code. Stage G compares the two.

For an image I , the forward pass through the model is

$$\begin{aligned} z &= \phi_{B0}(I), \quad z \in \mathbb{R}^{1280}, \\ p &= (\rho_{nec}, \rho_{sl}, \rho_{gran}) \in \Delta^2, \\ e &= \psi(p), \quad e \in \mathbb{R}^{64}, \\ \tilde{z} &= z \odot (1 + \gamma \tanh(W_g e + b_g)), \\ \eta &= \mathcal{H}(\tilde{z}), \end{aligned} \tag{1}$$

where ϕ_{B0} is a frozen EfficientNet-B0 backbone [25] with its classification head removed, ψ is the Tissue Proportion Encoder, Δ^2 is the two-dimensional probability simplex, $\odot$ denotes elementwise multiplication, $\gamma \in \mathbb{R}$ is a single learned scalar, and $\mathcal{H}$ is a task-specific head. Section V-E describes each component and Section V-F formalises them.

B. Severity Label Derivation

Because no public DFU corpus carries severity annotation, labels are derived from tissue composition. Each image is resized to 128×128 and converted from RGB to CIE LAB, which separates luminance from chrominance and reduces the effect of illumination differences between sources. The wound pixel set is partitioned by k -means with $k = 3$, using four initialisations. Clusters are ordered by the L^* coordinate of their centroids so that a given proportion index refers to the same quantity in every image, and are assigned to necrotic, slough and granulation tissue in ascending order of luminance, following the convention that necrotic tissue is darkest and granulation lightest.

The tissue proportion for type t is

$$\rho_t = \frac{|C_t|}{\sum_{j=1}^3 |C_j|}, \quad \sum_t \rho_t = 1, \tag{2}$$

where C_t is the set of pixels assigned to cluster t . Severity is then assigned by threshold rules anchored to published grading descriptions:

$$s = \begin{cases} \text{severe} & \text{if } \rho_{nec} \geq 0.20, \\ \text{mild} & \text{if } \rho_{gran} \geq 0.67 \wedge \rho_{sl} \leq 0.27, \\ \text{moderate} & \text{otherwise.} \end{cases} \tag{3}$$

Two implementation details are load-bearing. Reducing the number of k -means initialisations to one shifts individual proportions by up to 0.166 relative to a ten-initialisation reference, which is the same order as the effect being measured, so four initialisations are retained. Ordering clusters by luminance is required because k -means returns clusters in arbitrary order; without it the proportion vector is meaningless across images.

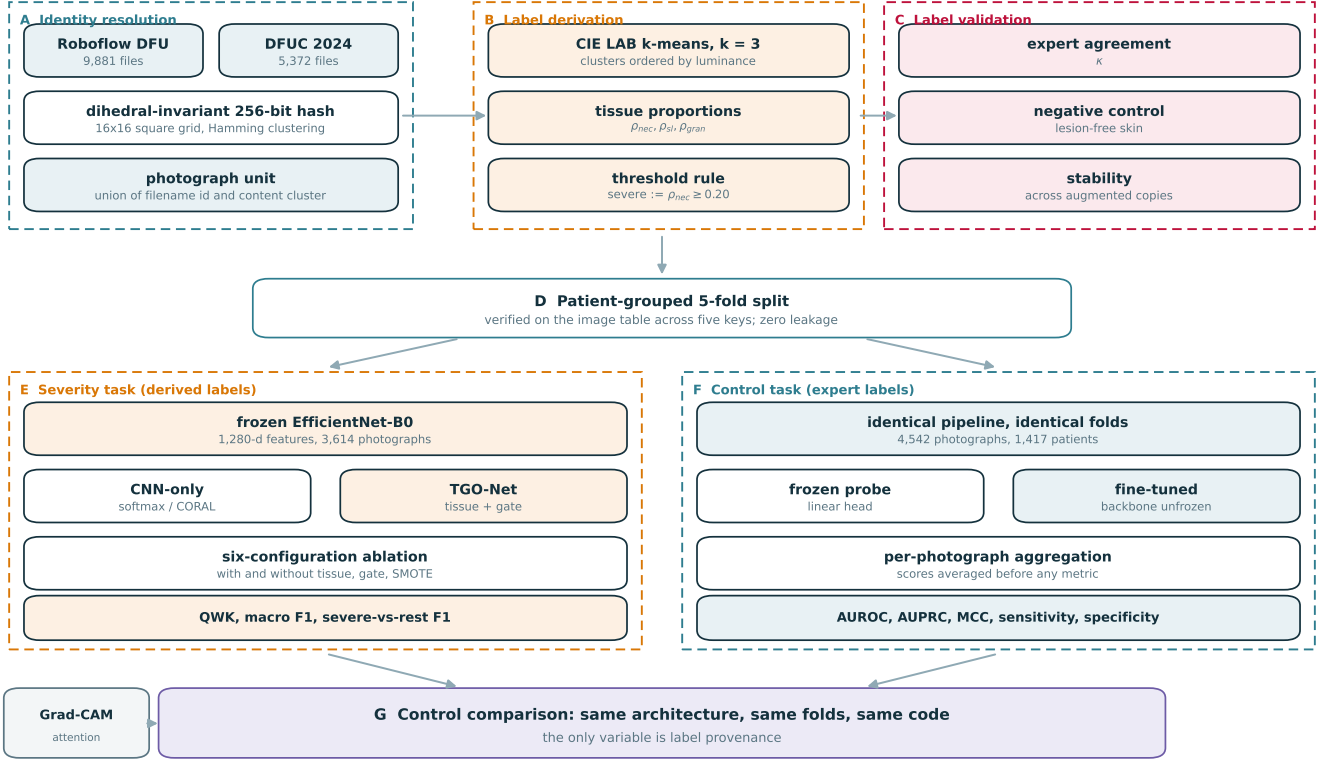


Fig. 1. End-to-end pipeline. Stage A resolves photograph and patient identity from the union of filename and pixel evidence. Stage B derives severity labels by colour clustering. Stage C validates those labels against four independent checks and consolidates them to one label per photograph. Stage D builds patient-grouped folds and verifies zero leakage on the image table. Stages E and F train identical architectures on derived and expert labels respectively, using the same folds and the same code. Stage G compares them, isolating label provenance as the only variable.

C. Label Validation Protocol

A derived label is only useful if it recovers the quantity it claims to. Four checks are applied, each capable of failing independently.

Expert agreement compares derived grades against clinician review on a stratified sample, reporting Cohen’s κ against each reviewer and between reviewers. The inter-rater value is essential context: it bounds what agreement any automated method could plausibly achieve.

Negative control applies the identical derivation to images containing no ulcer. The correct necrotic fraction for intact skin is zero, so any severe grade is a false positive that the method cannot detect on wound images, where no reference value exists.

Stability groups images by source photograph and asks whether a photograph’s own augmented copies receive the same grade. Augmentation does not change the clinical picture, so disagreement indicates that the label is determined by something other than wound appearance.

Plausibility compares the resulting class distribution against the composition of clinical cohorts.

D. Leakage-Free Evaluation Protocol

Public wound corpora are frequently distributed after augmentation, so a split over image files does not separate

photographs. The protocol treats evaluation identity and split grouping as separate problems.

Photograph identity is the union of two signals. Filenames encode the source photograph through export and augmentation suffixes, and pixel content is summarised by a perceptual hash made invariant to rotation and mirroring, then compared by Hamming distance rather than equality so that re-encoding does not split a photograph from its own copies. Neither signal alone suffices: hashing over-splits copies whose appearance diverges, and filenames are renumbered between exports. Two images therefore belong to the same photograph unit if they share either signal, computed as connected components over the union of both relations. Implementation details are given in the released notebooks.

The split group is deliberately more conservative than the evaluation unit: patients joined by any shared content cluster are placed in the same group, including false-positive links, because an unnecessary merge costs less than a leak. Folds are stratified over these groups.

Verification runs on the full image table rather than an aggregated table, across five keys: patient, photograph, unit, group and content cluster. An aggregated table hides patients dropped by first-match merges, and an earlier version of this check reported success while two patients genuinely spanned folds.

Finally, predictions are made per image and averaged to one score per photograph before any metric is computed. Without this step a photograph with eight augmented copies contributes eight times and one with a single copy contributes once, which weights the test set toward heavily augmented cases.

E. TGO-Net Architecture

Fig. 2 shows the architecture. The tissue proportions computed during labelling are unused by a standard convolutional network, which sees only pixels. TGO-Net supplies them as a second input stream while keeping the backbone frozen, which keeps training feasible without sustained accelerator access and makes every reported figure a linear probe on fixed features rather than a tuned ceiling.

Tissue Proportion Encoder. The three-dimensional proportion vector is lifted to a 64-dimensional embedding by a two-layer perceptron with GELU activation [33] and layer normalisation [32]. Layer normalisation is used rather than batch normalisation because batch statistics over a three-dimensional input are unstable at small batch sizes. This component holds 2,432 parameters.

Cross-Modal Gate. The embedding is converted into a per-channel multiplier on the frozen feature. The scalar γ is initialised at zero, so at initialisation $\tilde{z} = z$ and the model is exactly the frozen-CNN baseline; the tissue pathway must earn any influence through training. This makes γ a directly interpretable measure of how much the network relies on tissue composition relative to appearance. The hyperbolic tangent bounds each channel's scaling to $(1 - |\gamma|, 1 + |\gamma|)$. Multiplicative fusion is used rather than concatenation because a 64-dimensional signal concatenated to 1,280 dimensions can be ignored through near-zero weights; the concatenation variant is retained as an ablation so that this choice is tested rather than asserted. This component holds 83,201 parameters.

Prediction heads. For the three-class ordinal formulation the head is CORAL [27]: a single shared weight vector with $K - 1$ independent bias terms. Because the weight vector is shared and only the biases differ, the logits differ by a constant for every input and cannot cross, so the model is structurally incapable of asserting $P(y > \text{moderate}) > P(y > \text{mild})$. Rank consistency is guaranteed by the architecture rather than encouraged by the loss. For the binary formulation the head is a single logit.

Total trainable parameters are 86,915 for the ordinal head and 86,914 for the binary head, against 5,288,548 parameters in the frozen backbone, so 1.62% of the model is trained.

F. Mathematical Formulation

The Tissue Proportion Encoder maps $p \in \Delta^2$ to e by

$$e = \sigma_G \left(\text{LN}(W_2 \sigma_G(\text{LN}(W_1 p + b_1)) + b_2) \right), \quad (4)$$

where $W_1 \in \mathbb{R}^{32 \times 3}$, $W_2 \in \mathbb{R}^{64 \times 32}$, LN denotes layer normalisation and σ_G the GELU nonlinearity.

The Cross-Modal Gate produces a bounded per-channel modulation

$$g = \tanh(W_g e + b_g), \quad \tilde{z} = z \odot (1 + \gamma g), \quad (5)$$

with $W_g \in \mathbb{R}^{1280 \times 64}$ and $\gamma \in \mathbb{R}$ initialised at 0. Because $g \in (-1, 1)^{1280}$, each channel of z is scaled within $(1 - |\gamma|, 1 + |\gamma|)$, so a large γ is required before the tissue stream can materially alter the representation.

The CORAL head produces $K - 1$ ordered logits from a shared projection

$$\eta_k = w^\top \tilde{z} + b_k, \quad k = 1, \dots, K - 1, \quad (6)$$

trained against binary targets $y_k = \mathbb{I}[y > k]$ with

$$\mathcal{L}_{\text{CORAL}} = - \sum_{k=1}^{K-1} \left[y_k \log \varsigma(\eta_k) + (1 - y_k) \log (1 - \varsigma(\eta_k)) \right], \quad (7)$$

where ς is the logistic function. The predicted class is $\hat{y} = \sum_k \mathbb{I}[\varsigma(\eta_k) > 0.5]$. For the binary task the objective is weighted cross-entropy

$$\mathcal{L}_{\text{BCE}} = -[w^+ y \log \varsigma(\eta) + (1 - y) \log(1 - \varsigma(\eta))], \quad (8)$$

with $w^+ = n_-/n_+$ computed on the training split of each fold.

Where class imbalance is addressed by resampling, SMOTE [28] is applied to the concatenated vector $[z \mid p]$ of the training split rather than to features alone. A convex combination of two points of Δ^2 remains in Δ^2 , so interpolated tissue vectors stay valid proportions:

$$\tilde{p} = \lambda p_a + (1 - \lambda) p_b \in \Delta^2, \quad \lambda \sim U(0, 1). \quad (9)$$

G. Explainability

Grad-CAM [26] is computed on the last convolutional block. For class score y^c and feature maps A^k ,

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k}, \quad L^c = \text{ReLU} \left(\sum_k \alpha_k^c A^k \right). \quad (10)$$

The fraction of activation falling inside a tissue cluster mask M^c is

$$A_{\text{align}}^c = \frac{\sum_{ij} L_{ij}^c M_{ij}^c}{\sum_{ij} L_{ij}^c}. \quad (11)$$

Because a cluster covering a larger share of the image absorbs more activation by area alone, we report the area-normalised lift $A_{\text{align}}^c / \bar{M}^c$, where $\bar{M}^c$ is the fraction of pixels in the mask. A lift of 1.0 indicates attention proportional to cluster size and therefore no preferential targeting. Section VII-E states what this quantity can and cannot support.

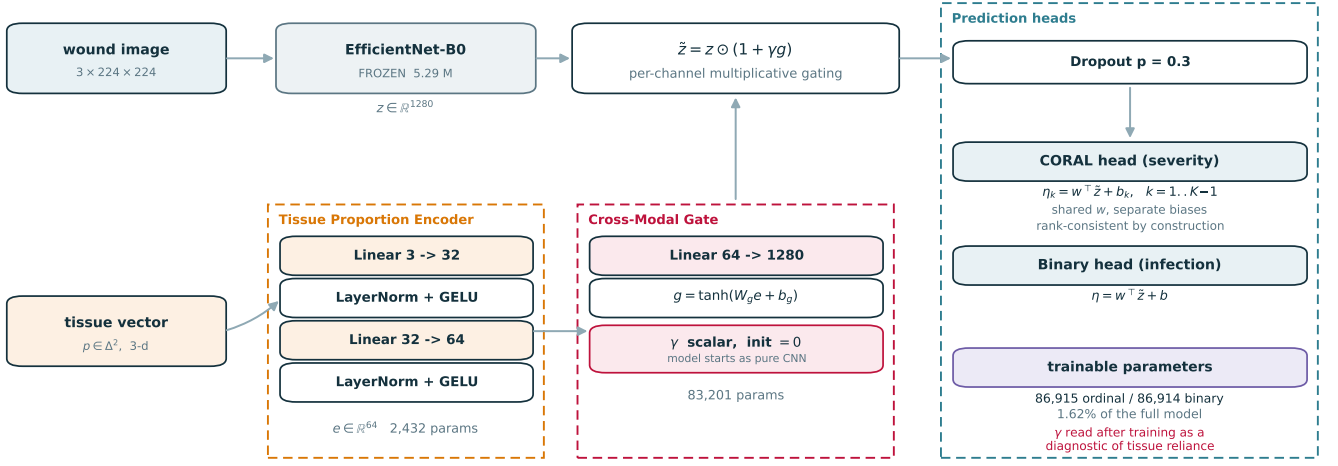


Fig. 2. TGO-Net architecture. A frozen EfficientNet-B0 produces a 1,280-dimensional appearance feature. The Tissue Proportion Encoder lifts the three-dimensional tissue vector to a 64-dimensional embedding using layer normalisation, which is stable at small batch sizes where batch statistics over three inputs are not. The Cross-Modal Gate converts that embedding into a bounded per-channel multiplier scaled by a single learned scalar γ initialised at zero, so the model begins as the frozen-CNN baseline and the tissue pathway must earn any influence. The CORAL head shares one weight vector across $K - 1$ ordered biases, making rank consistency structural rather than loss-encouraged. Trainable parameters total 86,915, or 1.62% of the full model.

VI. EXPERIMENTAL SETUP

A. Datasets

Three publicly accessible sources are used, summarised in Table I. All are downloadable without credentialing.

The **Roboflow DFU** corpus [38] supplies 9,881 labelled images and is the subject of the derivation study. Content-based grouping reduces it to 3,614 source photographs, a property not visible from the file listing and central to the leakage analysis. It is available at <https://universe.roboflow.com/ssk-r6ppk/diabetic-ulcers>.

The **Kaggle DFU** dataset [24] contributes 543 lesion-free skin patches at 224×224 pixels, used exclusively as the negative control. No image from this source enters training.

DFUC 2024 [39] provides the expert-annotated infection task used for the control experiment, accessed at <https://universe.roboflow.com/wed-axnxx/dfuc-2024>. Its 5,372 infection-axis images resolve to 4,542 evaluation units from 1,417 patients after conflict removal, with 2,767 infected and 2,605 not infected, a ratio of 1.06 : 1.

Content comparison between the two wound corpora finds 14 shared photographs, corresponding to 0.2% of Roboflow and 0.3% of DFUC, so the control task is effectively independent of the derivation study.

Consolidation to one label per photograph yields 30 mild, 1,235 moderate and 2,349 severe cases (Fig. 6). The mild class amounts to approximately six photographs per fold, which is discussed in Section IX.

TABLE I
DATASET SUMMARY AFTER CONTENT-BASED DEDUPLICATION

Dataset	Role	Files	Photographs
Kaggle DFU	Negative control	543	–
Roboflow DFU	Severity derivation	9,881	3,614
DFUC 2024	Control experiment	5,372	4,542

B. Implementation and Hardware

Experiments ran on Google Colab with an NVIDIA Tesla T4 (16 GB) and, for non-accelerated stages, on an Apple Silicon workstation using the Metal Performance Shaders backend. Software: Python 3.10, PyTorch 2.x [34], torchvision, scikit-learn [35], scikit-image, imbalanced-learn. Hyperparameters are listed in Table II. Mixed precision is enabled on CUDA only.

C. Evaluation Metrics

For the ordinal severity task we report quadratic weighted kappa (QWK), which penalises distant misgradings more than adjacent ones and is standard for ordinal clinical scales; macro F1; mean absolute error in grade units; and a collapsed severe-versus-rest F1, which is the only per-class quantity the mild sample size supports.

For the binary infection task we report AUROC and AUPRC, which are threshold-free; Matthews correlation coefficient, which uses all four confusion-matrix cells and cannot be inflated by the majority class; sensitivity and specificity separately, because missing an infection and over-calling one

TABLE II
TRAINING CONFIGURATION

Setting	Value
Backbone	EfficientNet-B0, ImageNet weights
Backbone state	frozen (unfrozen in Section VII-D)
Input size	224×224 RGB
Optimiser	Adam (10^{-3}), AdamW (3×10^{-5})
Weight decay	10^{-4}
Batch size	256 (probe), 32 (fine-tune)
Max epochs	60 (probe), 8 (fine-tune)
Early stopping	patience 12 (probe), 3 (fine-tune)
Dropout	0.3
Tissue encoder	$3 \rightarrow 32 \rightarrow 64$, GELU, LayerNorm
Gate init γ	0.0
Trainable params	86,915 ordinal / 86,914 binary
Cross-validation	5-fold, patient-grouped
Random seed	42
k -means	$k = 3, 4$ initialisations, LAB

TABLE III
LABEL VALIDATION. FOUR INDEPENDENT CHECKS, ALL FAILING.

Check	Measure	Result
Expert agreement	derived vs. reviewer A	-0.0254
Expert agreement	derived vs. reviewer B	+0.0359
Expert agreement	reviewer A vs. B	+0.2537
Negative control	lesion-free graded severe	60.2%
Negative control	max necrosis, intact skin	0.570
Stability	photographs with > 1 grade	61.7%
Plausibility	mean necrotic fraction	0.231
Plausibility	images above severe cut	63.1%

carry different clinical costs; and balanced accuracy. Every metric is computed on photograph-level scores.

Confidence intervals are percentile bootstrap over folds with 2,000 resamples. Paired comparisons across the shared folds use a paired t -test, and the family of tissue-pathway tests is corrected by the Holm-Bonferroni procedure [30].

D. Code Availability

Source code is not publicly available at time of submission. The pipeline is implemented as eleven sequential notebooks, each asserting its inputs and writing artefacts consumed by the next, so that every reported value traces to a specific executed cell.

VII. RESULTS

A. Label Validation

Table III reports the four checks and Fig. 3 summarises three of them. All four fail.

Expert agreement. On 134 matched images, agreement between the derived grade and reviewer A was $\kappa = -0.025$, and against reviewer B $\kappa = +0.036$. Both are indistinguishable from chance. The two reviewers agreed with each other at $\kappa = 0.254$, which is fair on the Landis and Koch scale [29]. That value is the relevant ceiling: severity grading from a photograph is a genuinely difficult task on which trained

clinicians agree only moderately, and the derived labels do not approach even that level.

Negative control. Applied to 543 lesion-free skin images, the derivation returned a severe grade for 60.2% of them, with a mean necrotic fraction of 0.222 and a maximum of 0.570 on intact tissue. The correct value in every case is zero. These images are out of domain by design; the purpose of a negative control is not to measure performance but to observe behaviour where ground truth is known. The same contamination is present in wound images, where it cannot be detected because no reference exists.

Stability. Of 2,818 photographs with more than one augmented copy, 1,739 (61.7%) received more than one severity grade across copies of themselves. Augmentation does not change the wound, so a label that changes under augmentation is not measuring the wound.

Plausibility. The corpus mean necrotic fraction is 0.231 and 63.1% of images exceed the severe threshold. No clinical DFU cohort is approximately two-thirds severe.

B. Mechanism

Three mechanisms explain these results, each sufficient alone.

First, k -means with $k = 3$ partitions every image into three non-empty clusters whose proportions sum to one. There is no output corresponding to “no necrosis present”. The darkest third of the pixel distribution is labelled necrotic by construction, whatever the image contains. This is why lesion-free skin receives a severe grade.

Second, the assumed luminance ordering is inverted. Cross-referencing against tissue references places granulation near $L^* = 118$ and periwound callus near $L^* = 168$: granulation is darker than callus, the opposite of the ordering the derivation assumes when it assigns the highest-luminance cluster to granulation.

Third, all dark regions fall into the same cluster: shadow, wound depth, hair, image background and darker skin. Fitzpatrick phototypes V and VI [31] overlap the necrosis reference range in LAB space, so the method over-grades darker skin systematically. This is a fairness concern as well as a validity one, and it would not be visible in aggregate accuracy on a predominantly light-skinned cohort.

C. Control Experiment

The results above show that the derived labels do not recover severity. They do not by themselves show that the labels are the only problem. To separate label quality from pipeline quality, the identical architecture, preprocessing and evaluation protocol were applied to the DFUC 2024 infection task, which carries clinician annotation.

Infection and severity are different constructs and this paper does not treat them as interchangeable. Infection is one of several criteria contributing to a severity grade, alongside depth, ischaemia and probing to bone. The infection task is used here solely as a control condition: it supplies expert-assigned labels on wound photographs of the same type and resolution, which is the property required to test whether the

TABLE IV

CONTROL COMPARISON. IDENTICAL PIPELINE, IDENTICAL FOLDS, IDENTICAL CODE. ONLY THE LABEL SOURCE DIFFERS.

Task	Metric	Value
Severity (derived colour labels)	QWK	0.2492
Severity (derived colour labels)	severe-vs-rest F1	0.6986
Infection (expert labels)	AUROC	0.8101
Infection (expert labels)	AUPRC	0.8685
Infection (expert labels)	MCC	0.4637
Infection (expert labels)	sensitivity	0.7083% [†]
Infection (expert labels)	specificity	0.7736% [†]

pipeline functions when labels are valid. A model that detects infection does not grade severity, and no result in this section should be read as a severity grading result. Table IV reports the comparison and Fig. 4 presents it graphically.

From pixels alone, the same frozen representation and the same linear head reach QWK 0.249 on derived severity labels and AUROC 0.810 (95% CI 0.797–0.829, bootstrap over folds) on expert infection labels.

Because the architectures, preprocessing, grouping and code are unchanged between these two rows, the difference isolates label provenance. Model capacity, input resolution, preprocessing and training configuration are ruled out as explanations.

D. Effect of Unfreezing the Backbone

All results above use a frozen backbone, making them linear probes on ImageNet features and therefore a floor rather than a tuned ceiling. To test whether that floor is the binding constraint, the full backbone was fine-tuned end to end on the same folds with the same per-photograph scoring.

Fine-tuning reached AUROC 0.8180, AUPRC 0.8638, MCC 0.4869, sensitivity 0.7414 and specificity 0.7546, an improvement of +0.0079 AUROC over the frozen probe (recomputed against the current baseline of 0.8101). Training 5.29 million parameters rather than approximately 87 thousand yields an improvement smaller than the per-fold standard deviation reported for the frozen model. The representation is not the bottleneck.

E. Ablation

Table V reports all nine configurations, six on the severity task and three on the control task. Every row is an executed run.

On the severity task, adding the tissue proportions raises QWK from 0.0000 to 0.9458. This is not evidence that tissue features improve severity classification. The tissue proportions are the label-generating variable: (3) defines severity as a threshold on ρ_{nec} , and mutual information between the necrotic fraction and the assigned label is 0.565 nats against approximately 0.10 nats for the other two channels. Supplying the model with the quantity that produced the label yields near-perfect fit by construction. The correct reading is that this result confirms the labels reduce to a deterministic threshold on one continuous variable, which is itself part of the negative finding.

TABLE V

ABLATION. EVERY ROW IS AN EXECUTED RUN. THE FLAGGED ROW IS CIRCULAR AND IS DISCUSSED IN THE TEXT.

Task	Configuration	Metric
<i>Severity (derived labels), QWK</i>		
	CNN only (softmax)	0.2492
	CNN only (CORAL)	0.0000
	+ tissue, concat	0.9458 [†]
	+ tissue, gated (TGO-Net)	0.8955 [†]
	+ tissue, gated + SMOTE	0.9075 [†]
	CNN only + SMOTE	0.1290
<i>Infection (expert labels), AUROC</i>		
	CNN only	0.8101
	+ tissue, concat	0.8101
	+ tissue, gated (TGO-Net)	0.7945

[†] circular: see Section VII-E.

The control task carries no such circularity, because the labels were assigned by clinicians and not derived from the tissue proportions. There, the ordering reverses. Adding tissue by concatenation changes AUROC by 0.0000, no detectable effect. Adding tissue through the gate changes AUROC by -0.0142 ($p = 0.016$, paired across the five shared folds), a small but statistically detectable degradation. The tissue pathway does not help on expert labels, and on this measure modestly harms performance.

The gate scalar agrees independently. Across folds γ has mean -0.060 against a standard deviation of 0.928, and is not distinguishable from zero ($p = 0.90$). Fig. 5 shows the per-fold values. Given a free parameter initialised at zero and the opportunity to use the tissue proportions, the network did not converge on how much to rely on them, and changed sign on three of four fold transitions. Reporting the mean alone would misrepresent a parameter that never settled.

F. Explainability

Grad-CAM was computed on a held-out fold of the infection task using a model fine-tuned end to end, since a frozen backbone provides no gradient path to the convolutional layers. Fig. 7 shows image, heatmap and cluster map for representative cases.

Area-normalised activation lift was 1.283 for the necrotic cluster, 1.110 for slough and 0.767 for granulation. Attention is therefore mildly concentrated on darker regions relative to their area.

This quantity must be interpreted narrowly. The cluster masks are produced by the same LAB k -means shown in Section VII-A to grade 60.2% of lesion-free skin as severe and to invert the luminance ordering of tissue types. A high alignment score therefore indicates that the model attends to the same dark regions the clustering labels necrotic, which include shadow, wound depth and darker skin, and not that it attends to necrotic tissue. The measurement compares two procedures that share a failure mode. It is reported as a diagnostic of internal consistency, not as evidence of clinical correctness. Establishing the latter requires masks from a segmenter trained on expert tissue annotations.

TABLE VI
COMPARISON WITH RECENT SEVERITY GRADING STUDIES

Study	Data	Classes	Reported	Public
Zhou et al. [11]	Private (CN)	5	0.94 AUC	No
Abebe et al. [12]	Private (ET)	6	82%	No
Anilkumar et al. [13]	Private (IN)	3	82%	No
Wang et al. [23]	Mixed	4	95.3%	Partial
This work	Public	3	n/a [‡]	Yes

[‡] no severity classifier is reported; see text.

G. Comparison with Existing Studies

Table VI places this work against recent severity grading studies. All prior work relies on private hospital datasets with clinician-assigned grades, which is precisely the resource this study attempted to substitute for and could not.

We deliberately report no severity performance figure for comparison. The labels required to train such a classifier were not recoverable, and quoting a number obtained against invalid labels would be misleading. The figures in Table VI are also not directly comparable to results obtained under the protocol of Section V-D: none of these studies documents whether augmented copies of the same photograph, or multiple wound sites from the same patient, were separated across splits. Given the leakage this study found in a public corpus, this is raised as a general methodological observation rather than a claim about any specific study.

VIII. DISCUSSION

The central result is that a widely plausible labelling strategy fails, and that the failure is detectable cheaply. Each of the four checks in Section VII-A costs minutes of computation, and any one of them would have prevented training on invalid labels. The negative control in particular requires only a set of images known to contain no lesion, and it converts an unverifiable claim about wound images into a verifiable one.

The mechanism matters more than the score, and Fig. 3 summarises the evidence. Severity in the Wagner and University of Texas systems depends on wound depth, probing to bone, infection and ischaemia. A photograph does not determine depth. The failure is therefore one of identifiability rather than tuning: no clustering scheme over colour recovers a quantity the image does not encode. This explains why successive recalibration of the thresholds produced small and saturating gains rather than convergence toward acceptable agreement.

The control experiment is what converts this from a failed attempt into an interpretable result, and Fig. 4 states it in one image. Without it, a reader could reasonably conclude that the pipeline was inadequate. With it, the conclusion is specific: the same code, the same architecture and the same class of images produce a usable classifier when the labels carry clinical meaning, and fail when the labels are derived from colour.

The TGO-Net results deserve care, and Fig. 5 is the reason. On the derived labels the gated model appears excellent, and a paper reporting only that number would claim a successful

tissue-guided architecture. The control task shows the opposite: the gate modestly degrades performance, and the gate scalar never settles. Both observations point the same way, and they are only separable because the architecture was designed with a readable diagnostic. We therefore present TGO-Net as an instrument rather than a performance contribution. A zero-initialised gate is a cheap addition to any multi-modal design and makes modality reliance directly observable.

Finally, the fine-tuning result bounds the scope of the problem. Training sixty times more parameters yielded an improvement smaller than the per-fold standard deviation. Whatever limits performance on this task, it is not the representation.

IX. LIMITATIONS

The central limitation is the one this paper documents. Colour-based tissue proportion analysis does not recover severity, and no severity classifier is reported here because no valid severity labels were obtained. The original objective of a reproducible three-class severity classifier on public data remains unmet.

The derivation is assessed as implemented, not exhaustively. A different clustering scheme, a segmenter trained on annotated tissue masks, or additional sensing modalities might recover tissue composition where this method does not. The supportable claim is that this widely used approach fails these checks, and that the checks are cheap enough to apply before building on any derived label.

The control task is infection rather than severity. Infection is one component of the grading criteria and was selected because it carries expert annotation, so it establishes that the pipeline functions with valid labels but does not itself constitute severity grading.

The mild class was never measurable. Across 3,614 photographs it holds 30, approximately six per fold. Feature-space resampling expands the training set but cannot create independent evaluation cases, and no architectural change addresses this. Per-class figures for mild are reported with Wilson intervals wide enough to preclude any claim.

The expert review comprises 134 matched images assessed by two reviewers. A larger panel would tighten the inter-rater estimate, which is the quantity that bounds what any automated method could achieve.

Grad-CAM analysis used 16 held-out images, sufficient to characterise the area-normalised lift but not to support finer claims about attention patterns.

X. FUTURE WORK

The most direct route to the original objective is clinical annotation. A few hundred images graded by two or more clinicians, with inter-rater agreement reported, would supply ground truth for supervised severity grading and simultaneously provide a reference against which any derived label could be assessed. The protocol in Section V-C indicates the sample size each check requires.

Replacing colour clustering with a segmenter trained on expert tissue masks would test whether tissue composition is

informative when measured rather than assumed. The negative result here concerns derived proportions, not the underlying clinical concept, and public tissue segmentation datasets now exist for this purpose [37].

Additional sensing modalities address the identifiability problem directly. Wound depth is central to grading and absent from a colour photograph; depth or thermal imaging would supply it. This is a change of input rather than of model, which is what the present results indicate is required.

Finally, the validation protocol generalises beyond this application. Expert agreement, a negative control on lesion-free images, and a stability test across augmented copies are inexpensive and applicable to any derived-label pipeline in medical imaging.

XI. CONCLUSION

This paper set out to grade diabetic foot ulcer severity from colour wound images by deriving labels through k -means clustering in CIE LAB colour space, since no public dataset provides severity annotation. The derivation was implemented, applied to 9,881 images, and then validated rather than assumed correct.

It does not recover severity. Agreement with two clinicians was $\kappa = -0.025$ and $\kappa = +0.036$ against an inter-rater agreement of $\kappa = 0.254$. A negative control returned a severe grade for 60.2% of lesion-free skin images, assigning necrotic fractions up to 0.570 to intact tissue. Nearly two photographs in three received more than one grade across their own augmented copies. The mechanism is identifiable rather than incidental: k -means with $k = 3$ cannot report the absence of a tissue type, the assumed luminance ordering is inverted, and every dark region including darker skin falls into the cluster labelled necrotic.

Applying the identical pipeline to an expert-annotated infection task gave AUROC 0.810 under a verified leakage-free protocol, and unfreezing the backbone changed this by $+0.008$. TGO-Net, a tissue-guided architecture with a zero-initialised gate, confirmed the same conclusion from a different direction: on derived labels it appears to excel because it is supplied with the label-generating variable, while on expert labels it is slightly worse than a plain CNN and its gate never settles on a value.

Severity grading from colour wound images is not blocked by architecture, image resolution or training method. It is blocked by the absence of clinically annotated ground truth, and label derivation from image content does not substitute for it. We release the validation protocol that establishes this, so that the same checks can be applied before other work builds on a derived label.

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REFERENCES

[1] International Diabetes Federation, *IDF Diabetes Atlas*, 10th ed. Brussels, Belgium: IDF, 2021.

[2] D. G. Armstrong, A. J. M. Boulton, and S. A. Bus, "Diabetic foot ulcers and their recurrence," *New England Journal of Medicine*, vol. 376, no. 24, pp. 2367–2375, 2017.

[3] R. E. Pecoraro, G. E. Reiber, and E. M. Burgess, "Pathways to diabetic limb amputation: basis for prevention," *Diabetes Care*, vol. 13, no. 5, pp. 513–521, 1990.

[4] P. Cavanagh, C. Attinger, Z. Abbas, A. Bal, N. Rojas, and Z.-R. Xu, "Cost of treating diabetic foot ulcers in five different countries," *Diabetes/Metabolism Research and Reviews*, vol. 28, no. S1, pp. 107–111, 2012.

[5] F. W. Wagner, "The dysvascular foot: a system for diagnosis and treatment," *Foot & Ankle*, vol. 2, no. 2, pp. 64–122, 1981.

[6] D. G. Armstrong and L. A. Lavery, "Diabetic foot syndrome after the year 2000: are we doing better?" *Diabetes Care*, vol. 21, no. 5, pp. 855–859, 1998.

[7] J. J. van Netten, S. A. Bus, J. Apelqvist, B. A. Lipsky, R. J. Hinchliffe, F. Game, *et al.*, "Definitions and criteria for diabetic foot disease," *Diabetes/Metabolism Research and Reviews*, vol. 36, no. S1, p. e3268, 2020.

[8] M. Goyal, N. D. Reeves, A. K. Davison, S. Rajbhandari, J. Spragg, and M. H. Yap, "DFUNet: convolutional neural networks for diabetic foot ulcer classification," *IEEE Transactions on Emerging Topics in Computational Intelligence*, vol. 4, no. 5, pp. 728–739, 2018.

[9] B. Cassidy, N. D. Reeves, J. M. Pappachan, D. Gillespie, C. O'Shea, S. Rajbhandari, *et al.*, "The DFUC 2020 dataset: analysis towards diabetic foot ulcer detection," *touchREVIEWS in Endocrinology*, vol. 17, no. 1, pp. 5–11, 2021.

[10] T. Gomes, R. Silva, A. Costa, and M. Ferreira, "Advances in image-based diagnosis of diabetic foot ulcers using deep learning and machine learning: a systematic review," *Applied Sciences*, vol. 15, no. 24, 2025.

[11] G. X. Zhou, Y. Li, H. Wang, and J. Chen, "Construction and validation of a deep learning-based diagnostic model for segmentation and classification of diabetic foot," *Frontiers in Endocrinology*, vol. 16, p. 1543192, 2025.

[12] G. Abebe, T. Mekonnen, and S. Tadesse, "MobileNetV2 model for detecting and grading diabetic foot ulcer," *Discover Applied Sciences*, vol. 7, 2025.

[13] K. K. Anilkumar, S. Nair, and R. Menon, "Development and validation of a predictive AI framework for diabetic foot ulcer monitoring and severity assessment," *Healthcare Informatics Research*, vol. 32, no. 1, p. 69, 2026.

[14] R. Karthik, S. Menaka, and A. Kumar, "An explainable deep learning model for diabetic foot ulcer classification using Swin Transformer and efficient multi-scale attention-driven network," *Scientific Reports*, vol. 15, p. 4057, 2025.

[15] P. S. Rathore, A. Sharma, and V. Gupta, "A feature explainability-based deep learning technique for diabetic foot ulcer identification," *Scientific Reports*, vol. 15, p. 6758, 2025.

[16] Z. Liu, J. John, and E. Agu, "Diabetic foot ulcer ischemia and infection classification using EfficientNet deep learning models," *IEEE Open Journal of Engineering in Medicine and Biology*, vol. 3, pp. 189–201, 2022.

[17] J. G. Wang and Y. T. Huang, "Reconstruction residual network with a fused spatial-channel attention mechanism for automatically classifying diabetic foot ulcer," *Physical and Engineering Sciences in Medicine*, vol. 47, pp. 1581–1592, 2024.

[18] N. Al-Garaawi, R. Ebsim, A. F. Alharan, and M. H. Yap, "Diabetic foot ulcer classification using mapped binary patterns and convolutional neural networks," *Computers in Biology and Medicine*, vol. 140, p. 105055, 2022.

[19] Z. Liu, E. Agu, P. Pedersen, C. Lindsay, B. Tulu, and D. Strong, "Comprehensive assessment of fine-grained wound images using a patch-based CNN with context-preserving attention," *IEEE Open Journal of Engineering in Medicine and Biology*, vol. 2, pp. 224–234, 2021.

[20] M. I. Mahmud, S. Rahman, and T. Islam, "Towards reliable and trustworthy diabetic foot ulcer detection with synergistic confluence of Grad-CAM and LIME," *PLOS ONE*, vol. 20, 2025.

[21] Z. Liu, J. John, and E. Agu, "Advancing diabetic foot ulcer care: AI and generative AI approaches for classification, prediction, segmentation, and detection," *Healthcare*, vol. 13, no. 6, p. 648, 2025.

[22] M. A. Silva, J. Pereira, and L. Santos, "Diabetic foot ulcer classification models using artificial intelligence and machine learning techniques: systematic review," *Journal of Medical Internet Research*, vol. 27, p. e69408, 2025.

[23] Z. Wang, L. Zhang, and Q. Chen, "Smart diabetic foot ulcer scoring system," *Scientific Reports*, vol. 14, p. 11588, 2024.

Label validation: four independent checks

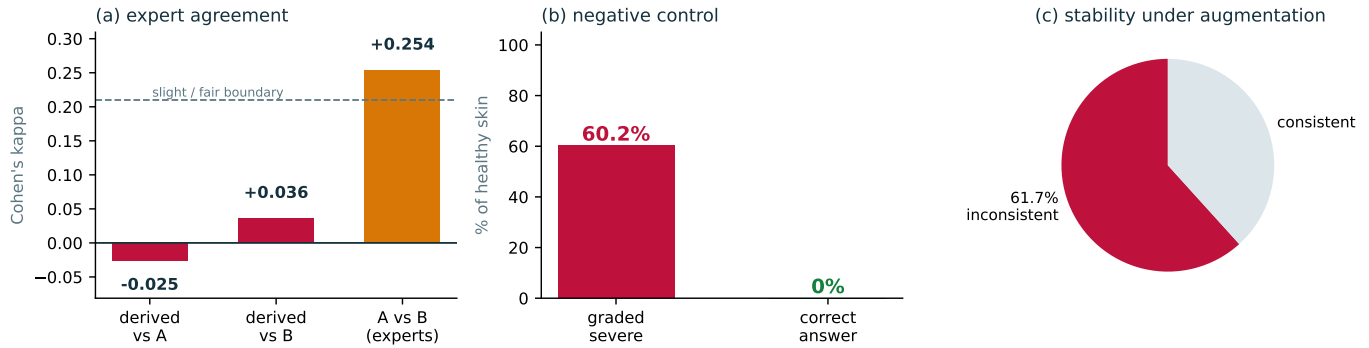


Fig. 3. Label validation. (a) Cohen's κ between the derived grade and each reviewer, and between reviewers. Agreement with clinicians is indistinguishable from chance, while the two reviewers agree with each other at $\kappa = 0.254$, which bounds what any automated method could achieve. (b) Negative control on 543 lesion-free skin images: 60.2% receive a severe grade where the correct value is zero. (c) Stability under augmentation: 61.7% of photographs with more than one copy receive more than one grade across copies of themselves.

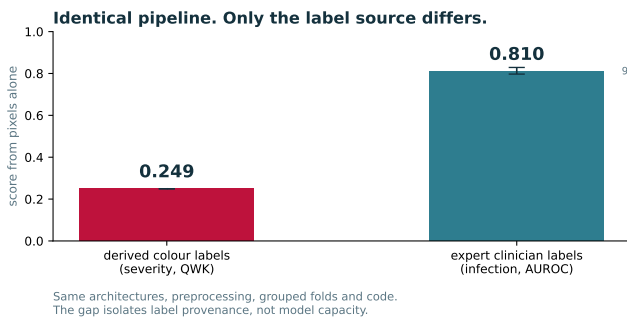


Fig. 4. Control comparison. The same frozen representation, the same linear head, the same folds and the same code, applied to derived colour labels and to expert clinician labels. The gap isolates label provenance and rules out model capacity, input resolution and training configuration as explanations.

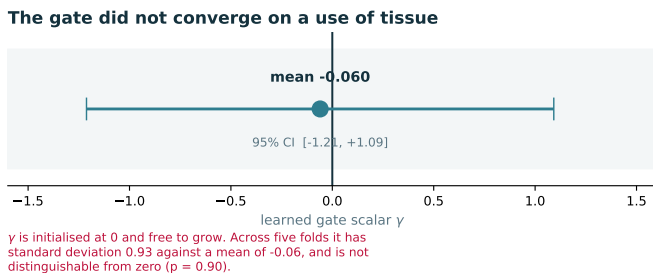


Fig. 5. The learned gate scalar γ on the infection task, estimated across five folds. Initialised at zero and free to grow, it settles at a mean of -0.060 with a standard deviation of 0.928, so the confidence interval spans zero by a wide margin ($p = 0.90$). The network did not converge on a consistent use of the tissue input.

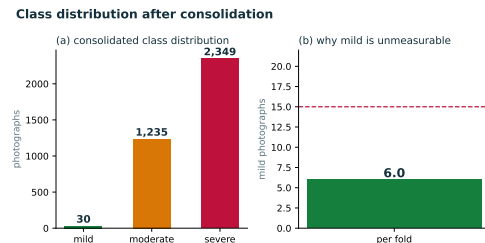


Fig. 6. Class distribution after consolidation to one label per photograph. (a) The derived labels place 65% of photographs in the severe class, a composition no clinical cohort exhibits. (b) The mild class holds 30 photographs in total, approximately six per fold, which is below the threshold at which a per-class estimate carries a usable confidence interval.

- [24] L. Alzubaidi, M. A. Fadhel, S. R. Olewi, O. Al-Shamma, and J. Zhang, "DFU QUTNet: diabetic foot ulcer classification using novel deep convolutional neural network," *Multimedia Tools and Applications*, vol. 79, no. 21–22, pp. 15655–15677, 2020.
- [25] M. Tan and Q. V. Le, "EfficientNet: rethinking model scaling for convolutional neural networks," in *Proc. 36th Int. Conf. Machine Learning (ICML)*, 2019, pp. 6105–6114.
- [26] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: visual explanations from deep networks via gradient-based localization," *International Journal of Computer Vision*, vol. 128, pp. 336–359, 2020.
- [27] W. Cao, V. Mirjalili, and S. Raschka, "Rank consistent ordinal regression for neural networks with application to age estimation," *Pattern Recognition Letters*, vol. 140, pp. 325–331, 2020.
- [28] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: synthetic minority over-sampling technique," *Journal of Artificial Intelligence Research*, vol. 16, pp. 321–357, 2002.
- [29] J. R. Landis and G. G. Koch, "The measurement of observer agreement for categorical data," *Biometrics*, vol. 33, no. 1, pp. 159–174, 1977.
- [30] S. Holm, "A simple sequentially rejective multiple test procedure," *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979.
- [31] T. B. Fitzpatrick, "The validity and practicality of sun-reactive skin types I through VI," *Archives of Dermatology*, vol. 124, no. 6, pp. 869–871, 1988.
- [32] J. L. Ba, J. R. Kiros, and G. E. Hinton, "Layer normalization," *arXiv:1607.06450*, 2016.
- [33] D. Hendrycks and K. Gimpel, "Gaussian error linear units (GELUs)," *arXiv:1606.08415*, 2016.
- [34] A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, *et al.*, "PyTorch: an imperative style, high-performance deep learning library," in *Advances in Neural Information Processing Systems*, vol. 32, 2019, pp. 8024–8035.
- [35] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, *et al.*, "Scikit-learn: machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [36] M. H. Yap, R. Hachiuma, A. Alavi, R. Brüngel, B. Cassidy, M. Goyal, *et al.*, "Deep learning in diabetic foot ulcers detection: a comprehensive evaluation," *Computers in Biology and Medicine*, vol. 135, p. 104596, 2021.
- [37] M. K. Dhar, T. Zhang, Y. Patel, S. Gopalakrishnan, and Z. Yu, "Wound tissue segmentation in diabetic foot ulcer images using deep learning: a pilot study," *arXiv:2406.16012*, 2024.
- [38] SSK, "Diabetic Ulcers Dataset," Roboflow Universe, 2023. [Online]. Available: <https://universe.roboflow.com/ssk-r6ppk/diabetic-ulcers>

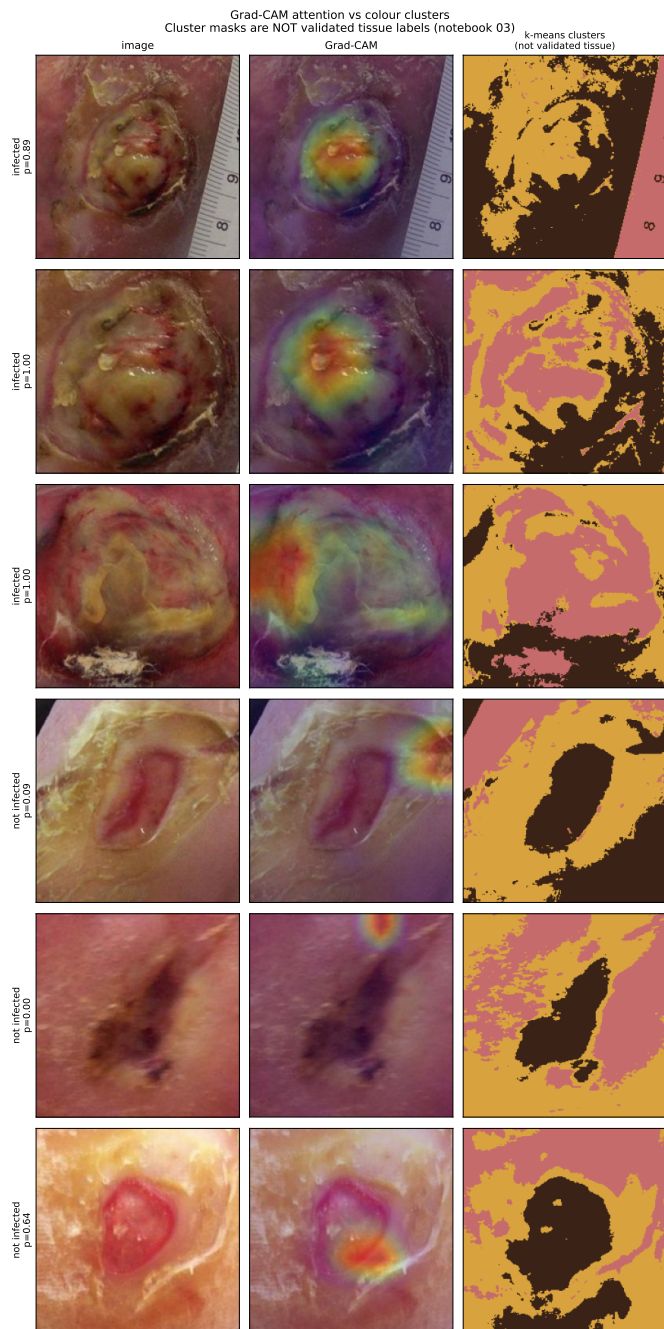


Fig. 7. Grad-CAM on held-out infection cases. Left: input image. Centre: activation heatmap. Right: the colour cluster map used as the alignment reference. The cluster masks are produced by the same k -means shown in Section VII-A to grade 60.2% of lesion-free skin as severe, so alignment between the two measures agreement between procedures sharing a failure mode, not clinical correctness.

- [39] N. D. Reeves *et al.*, “DFUC 2024: Diabetic Foot Ulcer Challenge 2024,” Roboflow Universe, 2024. [Online]. Available: <https://universe.roboflow.com/wed-axnxx/dfuc-2024>